

Boxing Robot

Ergonomics & Pad Placement — Design Reference

Target population: 150–190 cm stature | Stance: boxing defensive | Date: April 2026

1. Overview

This report documents the ergonomic analysis and pad-placement dimensions for a boxing training robot equipped with four target pads: one head pad and three body pads targeting the solar plexus (centre), liver (right), and spleen/floating-rib zone (left). All dimensions are derived from anatomical landmark data and adjusted for a standard boxing defensive stance (chin-tucked, slight knee bend).

The target population is adults with stature 150–190 cm, reflecting a mixed Southeast Asian and general adult demographic. Where Western anthropometric data (ANSUR) was used as a base, a proportional scaling note is provided.

2. Anatomical Target Locations

2.1 Solar Plexus (Celiac Plexus)

The solar plexus (celiac plexus) is a dense nerve network, not a surface organ. Anatomically it sits at the level of the first lumbar vertebra (L1), behind the stomach and in front of the crura of the diaphragm. The external surface landmark is the epigastrium — the pit of the stomach, just below the xiphoid process where the ribs converge at the midline.

The xiphoid-to-umbilicus distance averages approximately 18 cm in males and 17 cm in females. The solar plexus sits roughly in the upper third of this span, placing it ~3–6 cm inferior to the xiphoid.

2.2 Liver Target

The liver occupies the right upper quadrant, spanning from the 5th intercostal space (midclavicular line) superiorly down to the right costal margin inferiorly (approximately T8 to T12/L1 range). The boxing liver target is the right lower costal margin — ribs 9–10 on the right side, at approximately the T9–T10 vertebral level. This is meaningfully higher than the solar plexus, which sits at T10–L1.

Average liver span in the midclavicular line: 14.0 ± 1.7 cm (males 14.5 ± 1.6 cm, females 13.5 ± 1.7 cm).

2.3 Spleen / Left Floating-Rib Zone

The spleen occupies the left hypochondrium and lies slightly lower and more posterior than the liver. The left body target mirrors the liver side in height, but is positioned ~1–2 cm lower and further lateral due to the spleen's anatomy.

3. Population Baseline & Stance Correction

3.1 Anthropometric baseline

Primary data sourced from the ANSUR II dataset (US Army, n=6,068), which broadly represents Western adult population norms. Mean male stature ~175 cm. For a Singapore/Southeast Asian population, mean male stature is approximately 168–170 cm, representing a ~3–4% reduction. The 150–190 cm range used in this report spans the 5th–95th percentile for most adult populations.

3.2 Boxing stance correction

Standard anthropometric data assumes erect standing posture. A boxer in a defensive stance differs in two important ways:

- **Chin tuck (neck flexion):** Lowers the chin by approximately 5–10 cm.
- **Knee bend / crouch:** Lowers overall centre of mass by a further 5–10 cm.

Combined correction applied to chin height: **–15 cm** from erect posture. Solar plexus and body targets are corrected by **–10 cm** (the crouch effect; the solar plexus does not move relative to the spine, only the spine drops).

4. Stance-Adjusted Target Heights (from floor)

All heights below are in the boxing defensive stance, measured from floor to the centre of each target zone.

Stature	Chin (head pad centre)	Solar plexus centre	Liver centre	Spleen centre
150 cm	~112 cm	~78 cm	~85 cm	~83 cm
170 cm	~127 cm	~88 cm	~96 cm	~94 cm
190 cm	~143 cm	~98 cm	~107 cm	~105 cm

5. Pad Dimensions (as specified)

Pad	Height (length)	Notes
Head pad	230 mm	Chin at bottom edge of pad
Solar plexus	180 mm	Body pad, centre mount
Liver (right)	180 mm	Body pad, offset right
Spleen (left)	180 mm	Body pad, offset left

6. Fixed Gap Dimensions Between Pads

All gaps are measured edge-to-edge (bottom of upper pad to top of lower pad) in the boxing stance. Fixed design values represent the midpoint across the 150–190 cm target range.

6.1 Gap 1 — Bottom of head pad to top of solar plexus pad

Stature	Gap (cm)
150 cm	~34 cm

170 cm	~39 cm
190 cm	~45 cm
Fixed design value	~40 cm

6.2 Gap 2 — Bottom of liver/spleen pad to top of solar plexus pad

The liver/spleen pads are mounted higher than the solar plexus pad. Their bottom edges are therefore above the solar plexus pad's top edge, creating a small gap.

Stature	Liver pad bottom	Solar plexus top	Gap (cm)
150 cm	67 cm	78 cm	~11 cm
170 cm	78 cm	88 cm	~10 cm
190 cm	89 cm	98 cm	~9 cm
Fixed design value			~10 cm

7. Total Vertical Span

The total vertical span from the **bottom of the solar plexus pad** to the **bottom of the head pad** is calculated as:

$$\text{Total span} = \text{Gap 1} + \text{solar plexus pad height} = 40 \text{ cm} + 18 \text{ cm} = \mathbf{58 \text{ cm}}$$

This spans across stature range as follows:

Stature	Total span
150 cm	34 + 18 = 52 cm
170 cm	39 + 18 = 57 cm
190 cm	45 + 18 = 63 cm
Fixed design value	~58 cm

8. Lateral Offset & Centre-to-Centre Distance

The solar plexus pad sits at the body midline. The liver and spleen pads are laterally offset to the right and left respectively, aligned with the midclavicular line. The midclavicular line sits approximately 8–10 cm lateral to the midline in an average adult, scaling with shoulder width.

8.1 Lateral offset from midline

Stature	Lateral offset (midline to liver/spleen pad centre)
150 cm	~7–8 cm
170 cm	~8–10 cm
190 cm	~9–11 cm

Fixed design value	~9 cm
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8.2 Centre-to-centre diagonal distance

Using the fixed design values (6 cm vertical separation, 9 cm lateral offset), the straight-line centre-to-centre distance between the solar plexus pad and each body pad is:

$$\text{Distance} = \sqrt{(6^2 + 9^2)} = \sqrt{(36 + 81)} = \sqrt{117} \approx \mathbf{10.8 \text{ cm}}$$

Pad pair	Vertical sep.	Lateral offset	Diagonal (centre-to-centre)
Solar plexus ↔ Liver	~6 cm	~9 cm	~10–12 cm
Solar plexus ↔ Spleen	~5 cm	~9 cm	~9–11 cm

9. Fixed Design Summary

Recommended fixed dimensions for the boxing robot pad placement, optimised for the 150–190 cm stature range in boxing defensive stance:

Parameter	Fixed Value	Range across 150–190 cm
Gap 1: Head pad bottom → Solar plexus pad top	40 cm	34–45 cm
Gap 2: Liver/spleen pad bottom → Solar plexus pad top	10 cm	9–11 cm
Total span: Solar plexus pad bottom → Head pad bottom	58 cm	52–63 cm
Vertical sep.: Solar plexus centre → Liver centre	6 cm	4–8 cm
Lateral offset: Midline → Liver/spleen pad centre	9 cm	7–11 cm
Diagonal: Solar plexus centre ↔ Liver centre	~10.8 cm	10–12 cm

Note: The liver pad should be mounted **higher** than the solar plexus pad by ~5–8 cm (centre-to-centre vertically). Mounting all three body pads at the same height is anatomically incorrect.

10. Population Note

The base calculations used ANSUR II (US Army, mean male stature ~175 cm) as the primary anthropometric reference, cross-referenced with general anatomical literature. For a Singapore / Southeast Asian population (mean male stature ~168–170 cm, approximately 3–4% shorter):

Gap	Western baseline	Singapore-adjusted
Gap 1 (head → solar plexus)	~40 cm	~38–39 cm
Gap 2 (liver/spleen → solar plexus)	~10 cm	~9–10 cm
Total span	~58 cm	~55–56 cm

11. Sources & References

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- [12] Legends Boxing — Basic Boxing Defense Moves. <https://www.legendsboxing.com/post/basic-boxing-defense-moves> (Torso rotated ~45 degrees; chin tucked behind lead shoulder.)
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